ML PROBLEM FRAMING & RESPONSIBLE DATA CARD

Customer Churn — Assignment Report

Prepared from the supplied customer-churn-training.csv and responsible-data-card-template.md

Dataset facts at a glance

Item	Observed result
Rows × columns	12 × 7
Target balance	Not churned: 7 (58.3%); churned: 5 (41.7%)
Missing cells	0
Exact duplicate rows	0
Critical finding	plan_type perfectly separates the target in this sample; investigate possible leakage or sampling artefact
Decision	No production ML approval; exploratory work only

Scope note: this report distinguishes observed facts from proposals and unknowns. Missing documentation is not treated as proof of permission or validity.

Executive summary

The supplied file contains 12 customer records and seven columns. Five records are labelled churned (1; 41.7%) and seven are labelled not churned (0; 58.3%). There are no missing cells and no exact duplicate rows in this file.

Important limitation: the dataset is extremely small, and its provenance, collection period, sampling method, consent status, label definition, and intended prediction horizon are not supplied. The plan_type field perfectly separates the labels in this sample: all five Basic records are churned, while all three Pro and all four Standard records are not churned. This may be a sampling artefact, a proxy, or leakage. It must be investigated and must not be treated as a reliable business rule.

Conclusion: machine learning is not justified for production decisions with this dataset. Use a transparent, human-reviewed, non-ML retention triage process while collecting a larger, representative, time-stamped dataset and validating the label and feature timing. The included notebook is an auditable exploratory baseline, not a production model.

1. ML problem-framing memo

Business decision: decide which currently active customers should be offered a voluntary retention contact or support check-in. A prediction may prioritise a human outreach queue; it must not automatically cancel, downgrade, penalise, deny service, change prices, or restrict any customer.

Prediction target: churned = 1 versus not churned = 0. The CSV does not define what 'churned' means, when it is measured, or the event window. Before modelling, the business owner must define churn operationally (for example, account cancellation or non-renewal) and specify the observation date and a forward-looking prediction horizon. Do not assume these details from the column name.

Unit of observation: one customer row, apparently identified by customer_id. Confirm that there is exactly one row per customer per observation date and define how repeat customer snapshots will be handled.

Action window: not provided. Proposed design for approval: score customers at a fixed weekly snapshot and use the score to prioritise outreach during the following 7 days, predicting a clearly defined churn event in a separately agreed future horizon. This is a proposal, not a fact contained in the file.

Non-ML baseline: first compare with a majority-class baseline that predicts the most common class (not churned for this sample). Also consider a transparent operational triage queue based on recent inactivity and repeated support contacts, with thresholds approved by the business. Do not use plan_type as a churn rule: it perfectly maps to the label in this sample and may encode leakage or sample construction.

Go/no-go: no production ML deployment until provenance and consent are documented, churn timing is defined, temporal train/test data is available, plan_type separation is explained, sample size and representativeness improve, and performance/fairness are independently validated.

2. Dataset inventory and observed profile

File: customer-churn-training.csv. Observed shape: 12 rows × 7 columns. Fields: customer_id, tenure_months, support_tickets, monthly_spend_inr, last_login_days, plan_type, churned.

Target distribution: churned=0: 7 (58.3%); churned=1: 5 (41.7%).

Data completeness: 0 missing cells across all columns. Exact duplicate rows: 0. The file has no timestamp, source-system field, split indicator, consent field, demographic fields, or documented sampling metadata.

Numeric observed ranges: tenure_months 1–30; support_tickets 0–5; monthly_spend_inr ■499–■1,499; last_login_days 1–30. plan_type has Basic, Standard, and Pro. These ranges describe only the 12 supplied rows and are not valid business limits.

Observed plan/label table: Basic: 5 churned, 0 not churned; Standard: 0 churned, 4 not churned; Pro: 0 churned, 3 not churned. This complete separation is a critical investigation item, not proof that plan type causes churn.

3. Feature and target dictionary

customer_id — customer identifier. Exclude from model inputs; it is a direct identifier and may allow memorisation. Keep separately for controlled operational joining only, with access restrictions.

tenure_months — number of months of customer tenure. Confirm it is measured as of the prediction snapshot. It may be useful but can proxy customer age or acquisition cohort.

support_tickets — count of support tickets. Confirm the counting window and ensure tickets are recorded before the prediction snapshot. It may proxy service friction and may be affected by unequal support access.

monthly_spend_inr — monthly spend in INR. Confirm billing period, refunds/discount handling, currency and that the value is available before the snapshot. Spending can proxy socioeconomic status; avoid using it to deny service or offer unequal treatment.

last_login_days — days since last login. Confirm it is measured at the snapshot and not after churn. It may be a strong behavioural signal but can disadvantage customers who use offline/low-frequency channels or have accessibility constraints.

plan_type — Basic/Standard/Pro subscription category. Investigate whether it is assigned before the prediction date and whether it is associated with the label by design. It perfectly separates the labels in this sample, so exclude it from the first diagnostic baseline until the cause is understood; assess proxy and fairness risks before any later use.

churned — target label (1/0). It is not a feature. Obtain the authoritative definition, event date, label window, censoring rules and source system. Confirm that it is not derived from any feature measured after the prediction time.

4. Responsible Data Card (completed template)

Dataset purpose: limited exploratory analysis of customer-retention prioritisation. It may support research into whether a human outreach queue is worth testing. It must not make autonomous decisions about cancellation, eligibility, price, credit, service access, or customer value.

Provenance and permission: the supplied CSV contains no provenance statement, owner, creator, collection method, licence, consent record, lawful-basis statement, retention policy, or contact for data rights. Status is UNKNOWN, not assumed permitted. Obtain written confirmation from the data owner and privacy/legal team before any use beyond this assignment. Restrict access and do not attempt re-identification.

Population and representation: 12 rows only. The represented customer population, geography, time period, acquisition channels, and sampling method are unknown. No demographic or accessibility attributes are supplied, so representation and subgroup fairness cannot be assessed. The observed class split is 7/5. Plan-type separation indicates possible selection or

labelling artefact.

Features and target: see the feature dictionary above. Exclude customer_id. Treat churned only as the label. Confirm all predictors are available before the prediction timestamp. Treat plan_type as a leakage/proxy investigation priority. Spend, tenure, usage and support activity may indirectly proxy sensitive or socioeconomic circumstances.

Quality checks: zero missing cells; zero exact duplicate rows; 12 unique customer_id values; target counts 7 not churned and 5 churned. Ranges are plausible within this tiny file but require domain validation. There is no date or split column, so temporal separation and leakage-free train/test separation cannot be verified. Do not report a random split score as reliable evidence.

Risks and safeguards: sampling bias—collect a representative multi-period sample; label error—document churn definition and audit labels; leakage—enforce feature timestamps and exclude post-outcome fields; privacy—minimise identifiers, access-control, retention limits; proxy bias—review spend/plan/usage features and subgroup impacts; false positives—unnecessary or annoying contact; false negatives—missed retention opportunity; automation misuse—human review only, no adverse automated decisions.

Intended evaluation: compare majority-class and approved rule-based baselines before ML. For a validated temporal holdout, report confusion matrix, precision, recall, F1, specificity, PR-AUC (especially if churn prevalence is low), ROC-AUC as secondary, calibration/Brier score, and threshold-specific expected cost. Evaluate subgroup error rates and calibration only where lawful, appropriate, and adequately powered. Conduct false-positive/false-negative case review. Pre-register thresholds and costs before model training.

5. Baseline and evaluation plan

Baseline A — majority class: predict not churned for every customer because it is the majority label in this sample. On the full 12-row descriptive dataset this yields 7/12 = 58.3% accuracy and 0% recall for churn. It is intentionally simple and demonstrates why accuracy alone is inadequate.

Baseline B — transparent human triage: customer-success staff may review recent inactivity and repeated support contacts using business-approved thresholds, then decide whether a respectful check-in is appropriate. Thresholds are not learned from this tiny dataset. Record the reason for contact and allow customers to opt out.

Do not claim a validated score from this dataset. If a candidate rule is illustrated in the notebook, it is explicitly exploratory and in-sample; it cannot establish generalisation. Do not select thresholds using the same records and then report those records as independent test data.

Future evaluation: create time-ordered train, validation and final test periods, grouping all snapshots for a customer so the same customer cannot leak across splits. Use a final untouched future period. Compare with the same operational baseline. Report confidence intervals and sample counts. Tune threshold on validation only; freeze it before final testing.

Metric interpretation: precision answers how many contacted customers truly churned; recall answers how many churners were identified; false-positive rate captures unnecessary outreach; false-negative rate captures missed churners. Choose thresholds using the actual cost and capacity of outreach, not accuracy alone. Calibration checks whether predicted probabilities match observed event rates.

6. Error costs, human review and abstention

False positive (predicted churn, customer would stay): wasted staff time, unwanted contact, loss of trust, perceived pressure, and potentially inappropriate incentives. Mitigate with low-friction, respectful contact, frequency caps, opt-out, and review before contact.

False negative (predicted stay, customer churns): missed opportunity to resolve a service issue and potential revenue loss. Mitigate with accessible support channels that remain available to all customers, not only model-flagged customers.

Abstention: if required fields are missing, values are out of validated ranges, the customer is outside the validated population, or model confidence is below a pre-approved threshold, do not score; route to a human or the standard service process.

Human review: a trained customer-success reviewer checks the reason, recency, context and contact history before outreach. The reviewer can override the flag and must record a concise reason. Do not expose unnecessary customer data.

Safeguards: no automatic penalties, price changes, service restrictions, or cancellation actions. Provide a way to correct inaccurate data and opt out of marketing contact.

7. Monitoring, rollback and governance

Before launch, assign named owners for data, model, business operations, privacy and incident response. Maintain versioned dataset, code, feature definitions, thresholds, approvals and decision logs.

Monitor weekly during any controlled pilot: input missingness and range violations; population and plan mix; prediction/flag volume; outreach capacity; opt-out/complaint rate; observed churn prevalence; precision/recall when labels mature; calibration; subgroup error rates where permitted; and drift in feature distributions.

Trigger human review and pause scoring if provenance/permission is withdrawn or unclear, label definitions change, data pipeline timestamps fail, leakage is discovered, missingness/range violations exceed pre-set limits, drift is material, subgroup harm appears, complaints rise, or performance falls below the pre-approved floor.

Rollback: stop generating new scores; revert to the documented non-ML customer-service process; preserve audit logs; notify accountable owners; investigate root cause; correct data/model; revalidate on a fresh time-based holdout; and require formal approval before restart.

8. Risk register

R1 Data provenance/permission unknown — High — confirm owner, licence/consent/lawful basis and permitted purpose; block production use until resolved.

R2 Tiny and possibly unrepresentative sample — High — collect a larger multi-period sample and document sampling frame; no production generalisation.

R3 Target definition/timing unknown — High — define churn event, label window, observation timestamp and censoring rules; audit labels.

R4 Plan-type perfect separation — High — investigate generation/selection process and temporal ordering; exclude plan_type from initial modelling until resolved.

R5 Identifier memorisation/privacy — Medium — remove customer_id from features; restrict join key and access.

R6 Sensitive proxies in spend/tenure/usage — Medium/High — conduct feature necessity and fairness review; prohibit adverse decisions.

R7 False-positive outreach — Medium — human approval, frequency caps, opt-out and complaint monitoring.

R8 False-negative missed support — Medium — retain universal support access and monitor recall.

R9 Temporal leakage/train-test contamination — High — time-based split, customer grouping, feature timestamp audit and untouched future test.

R10 Automation misuse — High — prohibit autonomous adverse action; document permitted use and enforce human oversight.

R11 Drift and changing customer behaviour — Medium — monitor inputs and delayed outcomes; pause and revalidate on drift.

R12 Uncalibrated probabilities/threshold — Medium — calibrate on validation data, assess Brier score and choose threshold by documented costs.

9. Final recommendation and completion checklist

Recommendation: NOT APPROVED for production ML. APPROVED only for controlled educational exploration and data-quality investigation. Use the non-ML human-reviewed process while the data owner resolves provenance, consent, label definition, sampling and the plan_type anomaly.

Completion checklist: [x] decision and prediction target described with unknowns flagged; [x] observation unit and proposed action window documented; [x] non-ML baseline specified; [x] all template data-card sections completed; [x] features, target, leakage and proxies reviewed; [x] missingness, duplicates, ranges and class balance checked; [x] metric and error-cost plan specified; [x] abstention and human review designed; [x] monitoring and rollback conditions defined; [x] risk register provided; [x] reproducible baseline notebook supplied. [ ] external provenance/consent verification; [ ] confirmed churn definition and time horizon; [ ] representative temporal dataset; [ ] approved business costs and thresholds; [ ] independent fairness and performance validation. These last items cannot be completed from the supplied files.

Appendix A — Descriptive data checks

Column	Type	Missing	Unique	Observed range / values
customer_id	object	0	12	C001, C002, C003, C004, C005, C006, C007, C008, C009, C010, C011, C012
tenure_months	int64	0	12	1 to 30
support_tickets	int64	0	6	0 to 5

Column	Type	Missing	Unique	Observed range / values
monthly_spend_inr	int64	0	5	499 to 1499
last_login_days	int64	0	11	1 to 30
plan_type	object	0	3	Basic, Pro, Standard
churned	int64	0	2	0 to 1

Appendix B — Observed plan type by churn label

Plan type	Not churned (0)	Churned (1)
Basic	0	5
Pro	3	0
Standard	4	0

Appendix C — Submission notes

Before submission, add your name, roll number, course, faculty name and submission date if required. These details were not supplied and have not been invented.